

step 2 Assume that $E \in \mathcal{L}(\mathbb{R}^n)$ i.e., is Lebesgue measurable.

By monotonicity,

$$\sup \{ \mathcal{L}^n(K) : K \subset E, K \text{ compact} \} \leq \mathcal{L}^n(E).$$

We prove that

$$\mathcal{L}^n(E) \leq \sup \{ \mathcal{L}^n(K) : K \subset E, K \text{ compact} \}. \quad (6.8)$$

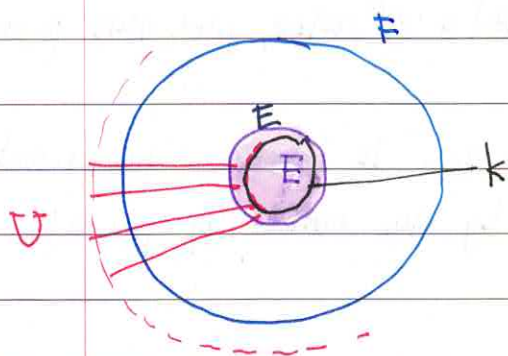
Suppose first that E is a bounded set, so that $\mathcal{L}^n(E) < \infty$.
Let $F \subset \mathbb{R}^n$ be a compact set that contains E . By step 1
for all $\varepsilon > 0$ there exists an open set

$$U \supset F \setminus E \quad (6.9)$$

$$\text{such that } \mathcal{L}^n(U) \leq \mathcal{L}^n(F \setminus E) + \varepsilon. \quad (6.9)$$

$$(\text{Note that } F \setminus E \in \mathcal{L}(\mathbb{R}^n)). \text{ Then } K := F \setminus U \quad (6.10)$$

$$\text{is a compact set and } K \subset E. \quad (6.11)$$



Indeed, if $x \in K$ then $x \in F$ and
 $x \notin U$. In particular, $x \notin F \setminus E$ (see (6.9))
and thus $x \in E$.

Therefore,

$$\mathcal{L}^n(E) = \mathcal{L}^n(F) - \mathcal{L}^n(F \setminus E)$$

$$(6.9) \leq \mathcal{L}^n(F) - \mathcal{L}^n(U) + \varepsilon$$

$$(6.10) \leq \mathcal{L}^n(K) + \varepsilon,$$

$$F \subseteq U \cup \underbrace{(F \setminus U)}_K$$

and, in view of (6.11), we conclude (6.8).

Assume now that E is unbounded, and define

$$E_k := E \cap B(0, k), \quad k \in \mathbb{N}.$$

Then $E_k \subseteq E_{k+1}$, $k \in \mathbb{N}$, they are $\mathcal{L}(\mathbb{R}^n)$ sets and by Prop 4.7

$$\mathcal{L}^n(E) = \lim_{k \rightarrow \infty} \mathcal{L}^n(E_k).$$

If $\mathcal{L}^N(E) = +\infty$ then $\mathcal{L}^N(E_k) \rightarrow +\infty$, and by the previous case we can find $u \rightarrow +\infty$

compact sets $K_k \subset E_k \subset E$ such that

$$\mathcal{L}^N(K_k) + 1 \geq \mathcal{L}^N(E_k).$$

Hence $\mathcal{L}^N(K_k) \rightarrow +\infty$ as $k \rightarrow +\infty$ and we conclude (6.8).

Finally, if E is unbounded and $\mathcal{L}^N(E) < +\infty$, given $\varepsilon > 0$ find $k \in \mathbb{N}$ such that

$$\mathcal{L}^N(E) \leq \mathcal{L}^N(E_k) + \frac{\varepsilon}{2}.$$

In turn, since E_k is a bounded $\mathcal{L}^N(\mathbb{R}^N)$ set, by the previous case we can find a compact set $K \subset E_k \subset E$ such that

$$\mathcal{L}^N(E_k) \leq \mathcal{L}^N(K) + \frac{\varepsilon}{2}.$$

We conclude that $\mathcal{L}^N(E) \leq \mathcal{L}^N(K) + \varepsilon$, and this proves (6.9).

Next, we will see that a set is Lebesgue measurable iff it can be approximated by the outside by open sets in the following sense:

Thm 6.14 A set $E \subset \mathbb{R}^N$ is Lebesgue measurable if and only if for every $\varepsilon > 0$ there exists an open set $U \supset E$ such that

$$(\mathcal{L}^N)^*(U \setminus E) < \varepsilon. \quad (6.12)$$

$$[\Rightarrow (\mathcal{L}^N)^*(U) \leq (\mathcal{L}^N)^*(E) + \varepsilon, \text{ see Def 6.5}]$$

Corollary 6.13* [Borel regularity of $(\mathcal{L}^N)^*$].

If $E \subseteq \mathbb{R}^N$ then $\exists B \in \beta(\mathbb{R}^N)$ s.t. $E \subseteq B$, $(\mathcal{L}^N)^*(E) = \mathcal{L}^N(B)$.

Proof By Thm 6.13 $\forall n \in \mathbb{N} \exists U_n$ open, $E \subseteq U_n$, $\mathcal{L}^N(U_n) \leq (\mathcal{L}^N)^*(E) + 1/n$.

Set $B := \bigcap_{n=1}^{\infty} U_n \in \beta(\mathbb{R}^N)$. Then $E \subseteq B$, $\mathcal{L}^N(B) \leq \mathcal{L}^N(U_n) \leq (\mathcal{L}^N)^*(E) + 1/n$ for all $n \in \mathbb{N}$, thus $\mathcal{L}^N(B) \leq (\mathcal{L}^N)^*(E)$.

Proof or see
Rmk. 6.8

Reason why
is Borel regular

[READ THE PROOF]

Proof $\Rightarrow$ Assume that E is Lebesgue measurable.

If $\mathcal{L}^N(E) < +\infty$, fix $\varepsilon > 0$.

By Thm 6.13 (6.41) there exists U open, $E \subset U$, and
 $\mathcal{L}^N(U) < \mathcal{L}^N(E) + \varepsilon$.

Since E is measurable, $\mathcal{L}^N(U) = \mathcal{L}^N(U \setminus E) + \mathcal{L}^N(U \cap E) = \mathcal{L}^N(U \setminus E) + \mathcal{L}^N(E)$
 $\Rightarrow (\mathcal{L}^N)^*(U \setminus E) = \mathcal{L}^N(U \setminus E) = \mathcal{L}^N(U) - \mathcal{L}^N(E) < \varepsilon$.

If $\mathcal{L}^N(E) = +\infty$, let $E_k := E \cap B(0, k)$ (measurable) and so
 Since $\mathcal{L}^N(E_k) < +\infty$, by the argument above for all $k \in \mathbb{N}$
 there is an open set $U_k \supset E_k$ such that
 $\mathcal{L}^N(U_k \setminus E_k) < \frac{\varepsilon}{2^k}$.

Then $U := \bigcup_{k=1}^{\infty} U_k$ is an open set containing E , and

$$\mathcal{L}^N(U \setminus E) = \mathcal{L}^N\left(\bigcup_{k=1}^{\infty} (U_k \setminus E)\right) \leq \sum_{k=1}^{\infty} \mathcal{L}^N(U_k \setminus E_k) < \varepsilon.$$

$\Leftarrow$ Suppose that (6.12) holds.

claim E is Lebesgue measurable, i.e., for all

$$A \subseteq \mathbb{R}^N$$

$$(\mathcal{L}^N)^*(A) \geq (\mathcal{L}^N)^*(A \cap E) + (\mathcal{L}^N)^*(A \setminus E).$$

Fix $\varepsilon > 0$, and choose an open set $U \supset E$ such that
 $(\mathcal{L}^N)^*(U \setminus E) < \varepsilon$.

Now $A \setminus E = (A \setminus U) \cup (A \cap (U \setminus E))$ so

$$(\mathcal{L}^N)^*(A \setminus E) + (\mathcal{L}^N)^*(A \cap E) \leq (\mathcal{L}^N)^*(A \setminus U) + (\mathcal{L}^N)^*(A \cap U) + (\mathcal{L}^N)^*(A \cap (U \setminus E))$$

$$\leq (\mathcal{L}^N)^*(A \setminus U) + (\mathcal{L}^N)^*(A \cap U) + \underbrace{(\mathcal{L}^N)^*(U \setminus E)}_{< \varepsilon}$$

$(U \text{ is measurable}) \leq (\mathcal{L}^N)^*(A) + \varepsilon$. This proves the claim. $\square$

The following theorem gives another characterization of Lebesgue measurable sets as those that can be "squeezed" between open and closed sets.

Thm 6.15 A set $E \subseteq \mathbb{R}^N$ is Lebesgue measurable if and only if for every $\varepsilon > 0$ there exist an open set U and a closed set F such that

$$F \subset E \subset U \quad \text{and} \quad \mathcal{L}^N(U \setminus F) < \varepsilon. \quad (6.13)$$

If, in addition, $\mathcal{L}^N(E) < \infty$ then F may be chosen to be compact.

[READ THE
PROOF!]

Proof ($\Rightarrow$) Assume that E is Lebesgue measurable.

Then E^c is Lebesgue measurable, and by Thm 6.14 there exist open sets $U \supset E$ and $H \supset (\mathbb{R}^N \setminus E)$ s.t.

$$\mathcal{L}^N(U \setminus E) < \frac{\varepsilon}{2}, \quad \mathcal{L}^N(H \setminus E^c) < \frac{\varepsilon}{2}.$$

Setting $F := \mathbb{R}^N \setminus H$, then F is closed,

$$F \subset E \subset U, \quad \text{and}$$

$$\begin{aligned} \mathcal{L}^N(U \setminus F) &= \mathcal{L}^N(U \setminus E) + \mathcal{L}^N(E \setminus F) \\ &= \mathcal{L}^N(U \setminus E) + \mathcal{L}^N(H \setminus E^c) \\ &< \varepsilon. \end{aligned}$$

[$\Leftarrow$] Assume that E satisfies (6.13)

Given $\varepsilon > 0$, find U open, F closed such that

$$F \subset E \subset U \quad \text{and} \quad \mathcal{L}^N(U \setminus F) < \varepsilon.$$

By monotonicity,

$$(\mathcal{L}^N)^*(U \setminus E) \leq \mathcal{L}^N(U \setminus F) < \varepsilon,$$

and by Thm 6.14 we conclude that E is Lebesgue measurable.

Finally, suppose that $\mathcal{L}^N(E) < +\infty$. Let $\varepsilon > 0$.

By Thm 6.13 (inner regularity) there exists a compact set $K \subset E$ such that

$$\mathcal{L}^N(E) < \mathcal{L}^N(K) + \varepsilon/2, \quad \text{need } \mathcal{L}^N(K) \leq \mathcal{L}^N(E) < \infty$$

so that $\mathcal{L}^N(E \setminus K) = \mathcal{L}^N(E) - \mathcal{L}^N(K) < \varepsilon/2$.

By outer regularity, let $U \supset E$ be open such that

$$\mathcal{L}^N(U) < \mathcal{L}^N(E) + \varepsilon/2.$$

Then $K \subset E \subset U$ and

$$\mathcal{L}^N(U \setminus K) = \mathcal{L}^N(U \setminus E) + \mathcal{L}^N(E \setminus K) < \varepsilon. \quad \blacksquare$$

Def 6.16 A G-delta in $\mathbb{R}^N$ is a countable intersection of open sets, and a F-sigma is a countable union of closed sets.

Note that G-delta and F-sigma sets are Borel sets.

Thm 6.17

A set $E \subset \mathbb{R}^N$ is Lebesgue measurable if and only if there exist a G-delta G and a F-sigma F such that $F \subset E \subset G$ and $\mathcal{L}^N(G \setminus F) = 0$.

Proof [$\Rightarrow$] Assume that E is Lebesgue measurable.

By Thm 6.15 for all $k \in \mathbb{N}$ find U_k open, F_k closed,

$$F_k \subset E \subset U_k, \quad \mathcal{L}^N(U_k \setminus F_k) \leq \frac{1}{k}.$$

Set $G := \bigcap_{k=1}^{\infty} U_k$, $F := \bigcup_{k=1}^{\infty} F_k$. Then $F \subset E \subset G$ and

for all k $\mathcal{L}^N(G \setminus F) \leq \mathcal{L}^N(G_k \setminus F_k) \leq 1/k$, hence $\mathcal{L}^N(G \setminus F) = 0$.

($\Leftarrow$) Note that $E \setminus F \subset G \setminus F$ with $G \setminus F$ Borel set of measure zero. Since $(\mathcal{L}^N)^*$ is a complete measure (see Example 4.10 (i)) it follows that $E \setminus F$ is measurable, with

$$E = (E \setminus F) \cup F, \text{ so measurable. } \blacksquare$$

Theorem 6.18

$E \subset \mathbb{R}^N$ is Lebesgue measurable iff there exist $B \in \beta(\mathbb{R}^N)$ and $A \in \mathcal{L}(\mathbb{R}^N)$ (Lebesgue measurable) with

$$E = B \cup A \text{ and } \mathcal{L}^N(A) = 0, \text{ with } A \subset H \in \beta(\mathbb{R}^N) \text{ with } \mathcal{L}^N(H) = 0.$$

Proof ($\Leftarrow$) obvious: $\mathcal{L}^N$ complete $\Rightarrow A \in \mathcal{L}(\mathbb{R}^N)$, so $E \in \mathcal{L}(\mathbb{R}^N)$.

($\Rightarrow$) Suppose that E is Lebesgue measurable. By

Thm 6.17 let G be a G -delta and F a F -sigma

$$\text{with } F \subset E \subset G, \quad \mathcal{L}^N(G \setminus F) = 0.$$

In particular $A := E \setminus F \in \mathcal{L}(\mathbb{R}^N)$ has measure zero,

By Thm 6.17, let H be a G -delta and F' be a F -sigma

$$\text{with } F' \subset A \subset H \text{ and } \mathcal{L}^N(H \setminus F') = 0$$

$$\text{Then } \mathcal{L}^N(H \setminus A) = \mathcal{L}^N(H) - \mathcal{L}^N(A) = 0$$

$$\text{and thus } \mathcal{L}^N(H) = 0.$$

We have $E = F \cup A$, where $F \in \beta(\mathbb{R}^N)$

$$\text{and } A \subset H \text{ with } H \in \beta(\mathbb{R}^N), \quad \mathcal{L}^N(H) = 0. \quad \blacksquare$$

Thm 6.19 The Lebesgue outer measure $(\mathcal{L}^N)^*$ is Radon.

Proof It follows from Def 6.7, Prop 3.13, and Thm 6.13. $\blacksquare$

Theorem 6.20 let $0 \leq s < \infty$.

(i) H^s_0 is a Borel measure on $\mathbb{R}^N$;

(ii) If $E \subseteq \mathbb{R}^N$ then

[Borel-regularity] $H^s_0(E) = \inf \{ H^s_0(B) : B \text{ Borel}, E \subset B \}$.

N.B. Example 4.10 (ii) (c): $s < N$, U open $\Rightarrow H^s_0(U) = +\infty$ because $\mathcal{L}^N(U) = H^N_0(U) > 0$.

(iii) If $E \subset \mathbb{R}^N$ is Borel and $H^s_0(E) < +\infty$ then

[inner regularity] $H^s_0(E) = \sup \{ H^s_0(K) : K \subset E, K \text{ compact} \}$

Proof. Exercise.

An extension of Thm 6.15 to metric spaces is the following:

Thm 6.21 let (X, d) be a metric space, and let μ be a finite Borel measure on X . Then for all $A \in \beta(X)$ and every $\varepsilon > 0$ there exist a closed set C and an open set U such that $C \subseteq A \subseteq U$ and $\mu(U \setminus C) < \varepsilon$.

Proof: let $\Lambda := \{ A \in \beta(X) : \forall \varepsilon > 0 \exists C \text{ closed}, U \text{ open}, C \subseteq A \subseteq U, \mu(U \setminus C) < \varepsilon \}$.

let $\Pi := \{ U \subset X : U \text{ is open} \}$.

clearly Π is a Π -system (see Def. 5.3). If we show

claim 1 $\Pi \subseteq \Lambda$

claim 2 Λ is a λ -system (Def 5.4) then

by Dynkin Thm 5.5 we conclude that

$$\beta(X) = \sigma(\Pi) \subseteq \Lambda.$$

[READ THE PROOF]

claim 1 let $U \subseteq X$ be open, $\varepsilon > 0$.

let $C_n := \{x \in U: \text{dist}(x, X \setminus U) \geq \frac{1}{n}\}$.

Then C_n is closed, $U = \bigcup_{n=1}^{\infty} C_n$ so

$$\phi = \bigcap_{n=1}^{\infty} (U|C_n) \text{ with } (U|C_n) \supseteq (U|C_{n+1})$$

and $\mu(U) \leq \mu(X) < +\infty$.

Hence, by Prop. 4.7 and because $\mu(X) < +\infty$,

$$0 = \lim_{n \rightarrow \infty} \mu(U|C_n) \text{ and choose } n \gg 1 \text{ s.t.}$$

$$\mu(U|C_n) < \varepsilon.$$

claim 2 Λ is a λ -system

o clearly $X \in \Lambda$, let $U_i = X$, $C_i = X$.

o let $A, B \in \Lambda$, $A \subseteq B$. ? $B \setminus A \in \Lambda$?

For $\varepsilon > 0$ let U_i open, C_i closed, $i=1,2$ with

$$C_1 \subset A \subset U_1, \quad C_2 \subset B \subset U_2, \quad \mu(U_i|C_i) < \varepsilon/2, \quad i=1,2.$$

Then

$$C_2|U_1 \subseteq B \setminus A \subseteq U_2|C_1,$$

$$C_2|U_1 \text{ closed, } U_2|C_1 \text{ open, and}$$

$$\begin{aligned} \mu((U_2|C_1) \setminus (C_2|U_1)) &= \mu(U_2 \cap C_1^c \cap C_2^c) \cup (U_2 \cap C_1^c \cap U_1) \\ &\leq \mu(U_2 \cap C_2^c) \cup (U_1 \cap C_1^c) \\ &\leq \mu(U_2 \setminus C_2) + \mu(U_1 \setminus C_1) < \varepsilon. \end{aligned}$$

o Now let $A_n \subseteq A_{n+1}$, $A_n \in \Lambda$. ? $\bigcup_{n=1}^{\infty} A_n \in \Lambda$?

$\forall n \in \mathbb{N}$ let $C_n \subseteq A_n \subseteq U_n$, C_n closed, U_n open,
 $\mu(A_n \setminus C_n) < \frac{\epsilon}{2^n}$. Write $A := \bigcup_{n=1}^{\infty} A_n$, $U := \bigcup_{n=1}^{\infty} U_n$.

Then $A \subseteq U$, U is open, and

$$\begin{aligned} \mu(U \setminus A) &= \mu\left(\bigcup_{n=1}^{\infty} (U_n \setminus A)\right) \leq \sum_{n=1}^{\infty} \mu(U_n \setminus A) \\ &\leq \sum_{n=1}^{\infty} \mu(U_n \setminus C_n) \end{aligned}$$

$$\leq \sum_{n=1}^{\infty} \mu(U_n \setminus C_n) < \epsilon. \quad (*)$$

Set $k_n := \bigcup_{i=1}^n C_i$, $k := \bigcup_{i=1}^{\infty} k_i$.

k_m closed,

$k_n \subseteq k_{n+1}$ and

$\mu(k) > \mu(k) - \epsilon = \lim_{n \rightarrow \infty} \mu(k_n)$. Hence $\exists m \in \mathbb{N}$ st.

$$\mu(k_m) > \mu(k) - \epsilon, \text{ i.e., } \mu(k \setminus k_m) < \epsilon.$$

Now,

$$\begin{aligned} \mu(A \setminus k_m) &= \mu(A \setminus k) + \mu(k \setminus k_m) \\ &= \mu\left(\bigcup_{n=1}^{\infty} (A_n \setminus k)\right) + \mu(k \setminus k_m) \\ &\leq \sum_{n=1}^{\infty} \mu(A_n \setminus k) + \epsilon \\ &\leq \sum_{n=1}^{\infty} \mu(A_n \setminus k_n) + \epsilon \\ &\leq \sum_{n=1}^{\infty} \mu(A_n \setminus C_n) + \epsilon \\ &\leq \sum_{n=1}^{\infty} \mu(U_n \setminus C_n) + \epsilon < 2\epsilon. \end{aligned} \quad (**)$$

$$(**) + (*) \Rightarrow \mu(A \setminus k) < 2\epsilon. \quad k \subseteq A \subseteq U$$

$$(i) \quad \sum_{i=1}^n (A - U) y_i = 0 \quad \text{for } y_i \in \mathbb{R}^n$$

$$(ii) \quad \sum_{i=1}^n (A - U) y_i = 0 \quad \text{for } y_i \in \mathbb{R}^n$$

$$(iii) \quad \sum_{i=1}^n (A - U) y_i = 0 \quad \text{for } y_i \in \mathbb{R}^n$$

$$(iv) \quad \sum_{i=1}^n (A - U) y_i = 0 \quad \text{for } y_i \in \mathbb{R}^n$$

$$(v) \quad \sum_{i=1}^n (A - U) y_i = 0 \quad \text{for } y_i \in \mathbb{R}^n$$

$$(vi) \quad \sum_{i=1}^n (A - U) y_i = 0 \quad \text{for } y_i \in \mathbb{R}^n$$

$$(vii) \quad \sum_{i=1}^n (A - U) y_i = 0 \quad \text{for } y_i \in \mathbb{R}^n$$

$$(viii) \quad \sum_{i=1}^n (A - U) y_i = 0 \quad \text{for } y_i \in \mathbb{R}^n$$

$$(ix) \quad \sum_{i=1}^n (A - U) y_i = 0 \quad \text{for } y_i \in \mathbb{R}^n$$

$$(x) \quad \sum_{i=1}^n (A - U) y_i = 0 \quad \text{for } y_i \in \mathbb{R}^n$$

$$(xi) \quad \sum_{i=1}^n (A - U) y_i = 0 \quad \text{for } y_i \in \mathbb{R}^n$$

$$(xii) \quad \sum_{i=1}^n (A - U) y_i = 0 \quad \text{for } y_i \in \mathbb{R}^n$$

7. Completion of a σ -algebra w.r.t. a measure

Let $(X, \mathcal{M}, \mu)$ be a measure space, and let

$$\mathcal{N} := \{ A \subseteq X : \exists H \in \mathcal{M} \text{ with } A \subseteq H \text{ and } \mu(H) = 0 \}$$

... the set of all μ -null subsets of X .

Recall (Def 3.8) that $\mathcal{M}$ is said to be complete w.r.t. μ if $\mathcal{N} \subseteq \mathcal{M}$.

Recall also that by Carathéodory Thm 4.10, the σ -algebra $\mathcal{M}^*$ constructed via an outer measure μ^* is complete.

Def 7.1 The completion $\mathcal{M}_\mu$ of $\mathcal{M}$ with respect to μ is defined as

$$\mathcal{M}_\mu := \{ B \cup A : B \in \mathcal{M}, A \in \mathcal{N} \}.$$

Proposition 7.2

The Lebesgue σ -algebra $\mathcal{L}(\mathbb{R}^N)$ is the completion of the Borel σ -algebra $\beta(\mathbb{R}^N)$ with respect to $\mathcal{L}^N$.

Proof This is the essence of Thm 6.18. ■

Lemma 7.3 $E \in \mathcal{M}_\mu$ if and only if there exist $F, G \in \mathcal{M}$ such that $F \subseteq E \subseteq G$ and $\mu(G \setminus F) = 0$.

Note: Compare with Thm 6.17.

Proof $[\Rightarrow]$ let $E \in \mathcal{M}_\mu$, with $E = B \cup A$, $B \in \mathcal{M}$, $A \in \mathcal{N}$.

let $H \in \mathcal{M}$ be such that $A \subseteq H$, $\mu(H) = 0$.

Set $F := B$, $G := B \cup H$. Then

$$F \subseteq E \subseteq G \quad \text{and} \quad \mu(G \setminus F) \leq \mu(H) = 0.$$

$[\Leftarrow]$ let $F, G \in \mathcal{M}$, $F \subseteq E \subseteq G$, $\mu(G \setminus F) = 0$.

Write $E = F \cup (E \setminus F)$. $E \setminus F$ is μ -null because

$$E \setminus F \subseteq G \setminus F \in \mathcal{M} \quad \text{with} \quad \mu(G \setminus F) = 0.$$

So $E \in \mathcal{M}_\mu$. ■

Thm 7.4 let $(X, \mathcal{M}, \mu)$ be a measure space, and

let $(X, \mathcal{M}_\mu, \bar{\mu})$ be its completion, where

$$\bar{\mu}(E) := \mu(B) \quad \text{if } E \in \mathcal{M}_\mu \text{ and } E = B \cup A, \quad B \in \mathcal{M}, \quad A \in \mathcal{N}.$$

(we'll see later that $\bar{\mu}$ is well defined, it does not depend on the decomposition B/A)

Then

- (i) $\mathcal{M}_\mu$ is a σ -algebra;
- (ii) $\bar{\mu}$ is a measure on $\mathcal{M}_\mu$, and $\bar{\mu}|_{\mathcal{M}} = \mu$;
- (iii) $\mathcal{M}_\mu$ is complete with respect to $\bar{\mu}$.

[READ THE
PROOF]

Proof (i) clearly $\emptyset \in \mathcal{M}_\mu$.

claim $E \in \mathcal{M}_\mu \Rightarrow X \setminus E \in \mathcal{M}_\mu$

By lemma 7.3 let $F, G \in \mathcal{M}$ with $F \subseteq E \subseteq G$, $\mu(G \setminus F) = 0$.

Then $X \setminus F, X \setminus G \in \mathcal{M}$, $X \setminus G \subseteq X \setminus E \subseteq X \setminus F$ and

$$\mu((X \setminus F) \setminus (X \setminus G)) = \mu(G \setminus F) = 0,$$

hence by lemma 7.3, $X \setminus E \in \mathcal{M}_\mu$.

claim $E_n \in \mathcal{M}_\mu \Rightarrow \bigcup_{n=1}^{\infty} E_n \in \mathcal{M}_\mu$.

let $E_n = B_n \cup A_n$, $B_n \in \mathcal{M}$, $A_n \in \mathcal{N}$. let $B := \bigcup_n B_n$, $A := \bigcup_n A_n$.
clearly $B \in \mathcal{M}$, $A \in \mathcal{N}$, and $\bigcup_{n=1}^{\infty} E_n = B \cup A$.

(ii) claim $\bar{\mu}$ is well defined.

let $E \in \mathcal{M}_\mu$ with $E = B_1 \cup A_1 = B_2 \cup A_2$, $B_1, B_2 \in \mathcal{M}$, $A_1, A_2 \in \mathcal{N}$.

let $H_1 \supseteq A_1$, $\mu(H_1) = 0$, $H_1 \in \mathcal{M}$,

$H_2 \supseteq A_2$, $\mu(H_2) = 0$, $H_2 \in \mathcal{M}$.

Then $\mu(B_1) \leq \mu(B_2 \cup H_2) \leq \mu(B_2) + \mu(H_2) = \mu(B_2)$,

and, similarly, $\mu(B_2) \leq \mu(B_1)$, so $\mu(B_1) = \mu(B_2)$ and

$\bar{\mu}$ is well defined.

clearly $\bar{\mu}(\emptyset) = 0$.

let $E_n \in \mathcal{M}_\mu$ be disjoint, $n \in \mathbb{N}$, $E_n = B_n \cup A_n$, $B_n \in \mathcal{M}$, $A_n \in \mathcal{N}$. Then

$$\bigcup_n E_n = B \cup A, \quad B := \bigcup_n B_n \in \mathcal{M}, \quad A := \bigcup_n A_n \in \mathcal{N}$$

and since $B_n \cap B_m = \emptyset$, for $n \neq m$, we have

$$\bar{\mu}\left(\bigcup_{n=1}^{\infty} E_n\right) = \mu(B) = \sum_{n=1}^{\infty} \mu(B_n) = \sum_{n=1}^{\infty} \bar{\mu}(E_n).$$

If $E \in \mathcal{M}$, then $E = E \cup \emptyset$, $\emptyset \in \mathcal{N}$, so that $\bar{\mu}(E) = \mu(E)$.

(iii) let A be a null set w.r.t. $\bar{\mu}$, i.e., there exists $H \in \mathcal{M}_\mu$ such that $A \subset H$, $\bar{\mu}(H) = 0$.

claim $A \in \mathcal{M}_\mu$

By lemma 7.3, let $F, G \in \mathcal{M}$ with $F \subseteq H \subseteq G$ and $\mu(G|F) = 0$.

Now $H = B \cup A'$, $B \in \mathcal{M}$, $A' \in \mathcal{M}$, $\mu(B) = 0 = \mu(H)$, $A' \subset H'$, $\mu(H') = 0$.
Hence $\mu(G) = \mu(F) \leq \mu(B \cup H') \leq \mu(B) + \mu(H') = 0$.

Therefore, $\emptyset \subseteq A \subseteq G$ with $G, \emptyset \in \mathcal{M}$, $\mu(G|\emptyset) = \mu(G) = 0$,
and by lemma 7.3, we deduce that $A \in \mathcal{M}_\mu$. ■

Remark 7.5

Note that taking the completion does not amount to including all sets of measure zero. Consider

$\mu := \mathbb{I}_0^N$, $X := \mathbb{R}^N$, $\mathcal{M} := \{ \emptyset, \mathbb{R}^N \}$. Then $\mathcal{M}_\mu = \mathcal{M}$.

8. MEASURABLE FUNCTIONS

Def 8.1 let $(X, \mathcal{M})$ and $(Y, \mathcal{N})$ be measurable spaces.

A function $f: X \rightarrow Y$ is said to be measurable if $f^{-1}(F) \in \mathcal{M}$ for every set $F \in \mathcal{N}$.

Lemma 8.2 let $(X, \mathcal{M})$ be a measurable space, and let Y be a non-empty set. let $f: X \rightarrow Y$.

Then $\mathcal{N}' := \{F \subseteq Y : f^{-1}(F) \in \mathcal{M}\}$ is a σ -algebra.

Proof (see Def. 3.4).

Clearly $f^{-1}(\emptyset) = \emptyset$, and so $\emptyset \in \mathcal{N}'$.

let $F \in \mathcal{N}'$. Then $f^{-1}(Y \setminus F) = f^{-1}(Y) \setminus f^{-1}(F) = X \setminus f^{-1}(F) \in \mathcal{M}$ and so $Y \setminus F \in \mathcal{N}'$.

Finally, if $F_n \in \mathcal{N}'$ then $f^{-1}(\bigcup_{n=1}^{\infty} F_n) = \bigcup_{n=1}^{\infty} f^{-1}(F_n) \in \mathcal{M}$, and we conclude that

$$\bigcup_{n=1}^{\infty} F_n \in \mathcal{N}'.$$

Corollary 8.3 let $(X, \mathcal{M})$ and $(Y, \mathcal{N})$ be measurable spaces, and let $\mathcal{N} = \sigma(\mathcal{E})$, $\mathcal{E} \subset \mathcal{P}(Y)$ (ie, $\sigma(\mathcal{E})$ is the smallest σ -algebra that contains $\mathcal{E}$, see Def. 3.8).

Then $f: X \rightarrow Y$ is measurable if and only if $f^{-1}(F) \in \mathcal{M}$ for all $F \in \mathcal{E}$. [it is enough to test measurability on generators of $\mathcal{N}$]

Proof $[\Rightarrow]$ clearly $\mathcal{E} \subseteq \mathcal{N}$, so this follows from Def 8.1.

$[\Leftarrow]$ let $\mathcal{N}' := \{F \subseteq Y : f^{-1}(F) \in \mathcal{M}\}$. Then

$\mathcal{E} \subseteq \mathcal{N}'$, hence (see Lemma 8.2) $\mathcal{N} = \sigma(\mathcal{E}) \subseteq \mathcal{N}'$. $\blacksquare$

Remark 8.4

(i) If $(X, \mathcal{M})$ is a measurable space and Y is a topological space, unless stated otherwise it is understood that the σ -algebra on Y is $\beta(Y)$ (σ -algebra of Borel sets).

By Corollary 8.3, $f: X \rightarrow Y$ is measurable if and only if $f^{-1}(U) \in \mathcal{M}$ for all $U \subset Y$ open.

(ii) If X and Y are topological spaces and we consider $(X, \beta(X))$, $(Y, \beta(Y))$, then a measurable function $f: X \rightarrow Y$ is called a Borel function.

(iii) Let $X: (\mathbb{R}^n, \mathcal{L}^n)$ (σ -algebra of Lebesgue measurable sets) and $(Y, \mathcal{M})$ be a measurable space. A measurable function $f: X \rightarrow Y$ is called Lebesgue measurable.

(iv) Let $(X, \mathcal{M})$ be a measurable space.

A function $f: X \rightarrow [-\infty, +\infty]$ is measurable

iff $\forall a \in \mathbb{R}, f^{-1}([-\infty, a]) \in \mathcal{M}$

iff $\forall a \in \mathbb{R}, f^{-1}((a, +\infty]) \in \mathcal{M}$

iff $\forall a \in \mathbb{R}, f^{-1}([a, +\infty]) \in \mathcal{M}$.

Indeed, this follows from (i), Corollary 8.3, and the fact that $\beta([-\infty, +\infty])$ is generated by all intervals of the form

- $[-\infty, a], a \in \mathbb{R},$
- or
- $(a, +\infty], a \in \mathbb{R},$
- or
- $[a, +\infty], a \in \mathbb{R}.$

(iv) If X and Y are topological spaces and $f: X \rightarrow Y$ is continuous, then f is Borel measurable.

In fact, by (i) we consider $(X, \beta(X))$ and $(Y, \beta(Y))$, and by (ii) f is Borel measurable if it is measurable w.r.t. these Borel σ -algebras.

By (i) (and Corollary 8.3) this follows from the fact that if $U \subset Y$ is open then $f^{-1}(U) \subset X$ is open, hence Borel.

In particular, $\forall F \in \beta(Y)$, $f^{-1}(F) \in \beta(X)$.

[see Example 4.12, Fact #3]

end proof #1

Examples 8.5

(i) let $f: \mathbb{R} \rightarrow \mathbb{R}$ be an increasing function.

We claim that f is Borel measurable (see Remark 8.4 (ii)).

In view of Corollary 8.3, it suffices to show that

for all $a \in \mathbb{R}$, $f^{-1}((a, +\infty)) \in \mathcal{B}(\mathbb{R})$.

Since f is increasing, if $b := \inf f^{-1}((a, +\infty))$ then

either $f^{-1}((a, +\infty)) = (b, +\infty) \in \mathcal{B}(\mathbb{R})$

or $f^{-1}((a, +\infty)) = [b, +\infty) \in \mathcal{B}(\mathbb{R})$.

(ii) let $(X, \mathcal{M})$ be a measurable space, and let $E \subseteq X$.

The characteristic function of E is defined by

$$\chi_E: X \rightarrow \mathbb{R}, \quad \chi_E(x) = \begin{cases} 1 & \text{if } x \in E, \\ 0 & \text{if } x \notin E. \end{cases}$$

$$\text{If } B \subseteq \mathbb{R} \text{ then } \chi_E^{-1}(B) = \begin{cases} E & \text{if } 1 \in B, 0 \notin B, \\ X \setminus E & \text{if } 0 \in B, 1 \notin B, \\ X & \text{if } 0 \in B, 1 \in B, \\ \emptyset & \text{if } 0 \notin B, 1 \notin B. \end{cases}$$

Thus χ_E is measurable iff $E \in \mathcal{M}$.

Proposition 8.6

let $(X, \mathcal{M})$, $(Y, \mathcal{N})$, $(Z, \mathcal{D})$ be measurable spaces, let $f: X \rightarrow Y$ and $g: Y \rightarrow Z$ measurable functions.

Then $g \circ f: X \rightarrow Z$ is measurable.

Proof If $F \in \mathcal{D}$ then $g^{-1}(F) \in \mathcal{N}$ and so $(g \circ f)^{-1}(F) = f^{-1}(g^{-1}(F)) \in \mathcal{M}$. ■

Remark 8.7

Read (i) If $(X, \mathcal{M})$ is a measurable space, $f: X \rightarrow \mathbb{R}^d$ is measurable and $g: \mathbb{R}^d \rightarrow \mathbb{R}^d$ is Borel measurable, then $g \circ f$ is measurable.

$f: X \rightarrow \mathbb{R}^d$ meas.
 $\downarrow \beta(\mathbb{R}^d)$

$g: \mathbb{R}^d \rightarrow \mathbb{R}^d$ Borel meas.
 $\downarrow \beta(\mathbb{R}^d)$

$\downarrow$
 $g \circ f$ meas.

This follows from Example 8.4 (i), for which for $f: X \rightarrow \mathbb{R}^d$ we consider the target as $(\mathbb{R}^d, \beta(\mathbb{R}^d))$.

Also, with $g: \mathbb{R}^d \rightarrow \mathbb{R}^d$ we take its target as $(\mathbb{R}^d, \beta(\mathbb{R}^d))$.

Therefore, if $B \in \beta(\mathbb{R}^d)$ then $g^{-1}(B) \in \beta(\mathbb{R}^d)$ (g is Borel meas.)
 $\Rightarrow f^{-1}(g^{-1}(B)) \in \mathcal{M}$, i.e., $(g \circ f)^{-1}(B) \in \mathcal{M}$.

(ii) let $f, g: \mathbb{R} \rightarrow \mathbb{R}$ be 'such that g is Lebesgue measurable (i.e., $g: (\mathbb{R}, \mathcal{L}) \rightarrow (\mathbb{R}, \beta(\mathbb{R}))$) and f is Borel measurable (i.e., $f: (\mathbb{R}, \beta(\mathbb{R})) \rightarrow (\mathbb{R}, \beta(\mathbb{R}))$).

$f: \mathbb{R} \rightarrow \mathbb{R}$ Borel meas.
 $\downarrow \beta(\mathbb{R})$

$g: \mathbb{R} \rightarrow \mathbb{R}$ Lebesgue meas.
 $\downarrow \beta(\mathbb{R})$

$\downarrow$
 $g \circ f$ meas.

Then $g \circ f$ may fail to be measurable,
 $g \circ f: (\mathbb{R}, \beta(\mathbb{R})) \rightarrow (\mathbb{R}, \beta(\mathbb{R}))$.

Indeed, by Example 4.12 there exists a set $F \subseteq \mathbb{R}$ that is Lebesgue measurable and not Borel measurable, with $F = g^{-1}(E)$, g continuous, g^{-1} continuous, and E is not Lebesgue measurable.